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Research Statement

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Learning and Feedback Control for Dexterous Manipulation

My research concerns how robots should select and execute manipulation actions when contact properties are uncertain and learned behavior changes with execution conditions. I develop grasp representations, state estimators, feedback controllers, and policy evaluations, and implement the software needed to study them in simulation and on physical arms and dexterous hands. I aim to develop manipulation systems that use perception and contact feedback to decide when to continue an action, revise a grasp, or seek additional information.

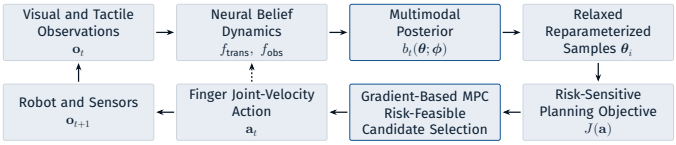


Figure 1. VNB-Grasp: Belief and Feedback Control
Visual/tactile observations update a multimodal belief for risk-sensitive action selection.

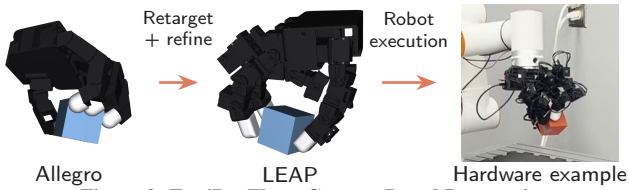


Figure 2. EquiDexFlow: Contact-Based Retargeting
Predicted contacts support retargeting from Allegro to LEAP and hardware execution.

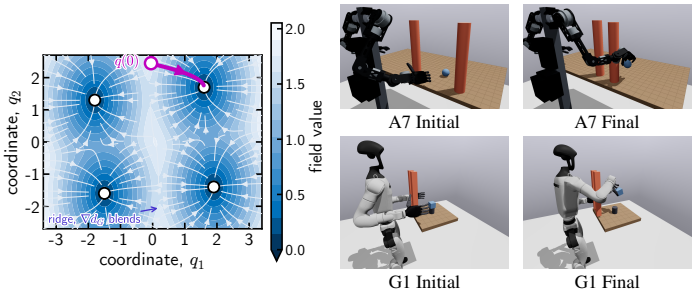


Figure 3. GDF: Field-Guided Grasp Execution
Smooth configuration-space field with initial/final A7 and G1 grasp-execution states in simulation.

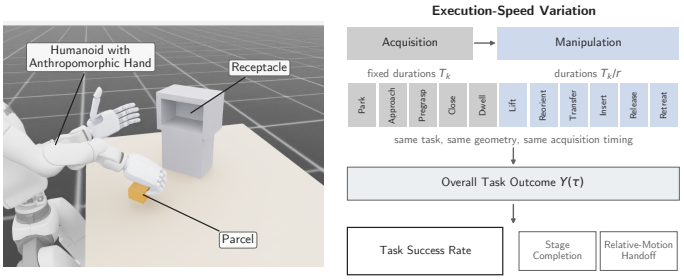


Figure 4. ParcelStow: Controlled Timing Evaluation
Acquisition timing stays fixed while manipulation durations vary; outcomes and task stages are evaluated separately.

Research Contributions

Physics-based reward design and completed reinforcement learning. I trained PPO in 1,024 parallel Isaac Lab environments using grasp-distance-field rewards for a Unitree G1 with a dexterous hand. The policy reached the pregrasp pose in 64/64 evaluation episodes versus 0/64 with palm-distance rewards. The result measures pregrasp reaching, rather than grasp-and-lift completion. My published CDC 2025 work [7] develops CVaR-regularized distributional RL with quantile regression and a contraction analysis. Across 100 MuJoCo reach–avoid episodes, the method reached 448 goals versus 329 and 239 for risk-neutral baselines, with similar constraint-violation costs.

Representing contact uncertainty for grasp selection and control. Object pose, friction, and contact behavior cannot generally be determined from a single observation. My IROS 2026 work, VNB-Grasp [1] (Fig. 1), represents pose and latent contact parameters with an eight-component Gaussian mixture. Relaxed sampling supports pathwise gradients of a smooth conditional value-at-risk (CVaR) objective for action optimization. The representation retains separated hypotheses instead of collapsing uncertainty into a single Gaussian. I evaluated the method across 72 MuJoCo conditions per method, varying friction regime, risk level, randomization, and object. In physical experiments across three objects and four planar pose offsets, VNB-MPC and Gaussian MPC each succeeded in 12 of 12 trials; median episode time was 11.5 s for VNB-MPC and 14.2 s for Gaussian MPC. The experiments distinguish the effects of the belief and controller from the hardware task's success rate.

My complementary work on FIRMGrasp [2] studies grasp selection under friction uncertainty through a CVaR wrench-space margin and a probabilistic force-closure certificate. This formulation makes friction sensitivity part of grasp evaluation. FIRMGrasp and VNB-Grasp examine uncertainty before execution, when selecting a grasp, and during execution, when observations can change the estimated contact conditions.

Belief-informed control under parameter uncertainty. In Belief-Informed Hybrid Control [6], I developed a dual-control algorithm whose two-step lookahead updates the parameter belief from predicted observations before evaluating the next action. An expected-decrease constraint and a bound on cumulative slack permit informative deviations from the target set. Under the stated prediction and feasibility assumptions, the method guarantees almost-sure target-set convergence at decision times. Simulated bimanual assembly experiments examine how contact feedback changes action admissibility under friction uncertainty.

Feedback execution of feasible grasps. In Grasp Distance Fields [3] (Fig. 3), I use a configuration-space distance field over feasible pregrasp configurations to guide feedback grasp execution. A control-barrier-function/control-Lyapunov-function quadratic program combines progress toward the grasp with collision and workspace constraints. In simulated grasp-and-lift experiments, the controller lifted 46 of 50 objects, correctly rejecting the remaining four grasps as insufficiently stable. The median QP solve time was 0.09 ms. Drake experiments additionally examine reactive execution with changing obstacles and multiple robot models. These simulation experiments examine execution across arm–hand and humanoid models.

Controlled evaluation of imitation policies. I built ParcelStow [4] (Fig. 4) to investigate whether nominal imitation performance persists when task execution timing changes. In Isaac Sim and Isaac Lab, I authored manipulation environments and generated 970K expert states from 297 demonstrations, then trained ACT, Diffusion Policy, and DAgger. The insertion benchmark compares policies with a scripted expert under matched execution speeds and initial conditions. At twice nominal speed, ACT succeeded in 53% of trials compared with 84% for the expert. Most ACT failures occurred during insertion. Stage-level evaluation therefore identifies a precision-manipulation limitation that aggregate nominal success does not reveal. The benchmark supplies a basis for testing whether additional demonstrations, different policy inputs, or feedback corrections address the observed failures.

Learned grasp generation and experimental software. In EquiDexFlow [5] (Fig. 2), I developed a flow-matching grasp model predicting wrist pose, joint configuration, fingertip contacts, and forces, trained on 8,100 grasps across 81 objects. I retargeted predicted contacts to a physical LEAP Hand and demonstrated ten successful pick-and-hold executions across six objects. I also implement ROS 2 hand interfaces, tactile acquisition, VR teleoperation, and motion-planning software. My RealHand L6 software includes a C++ SocketCAN driver, joint and tactile interfaces, contact-gated control, and MoveIt 2 configuration. I verified communication, feedback, and trajectory execution on the physical hand. I fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing. Each of three training seeds completed 136/200 tasks versus 129/200 without supervision, recovering seven failed executions without disrupting nominal successes. I also developed a depth-to-trajectory pipeline combining RANSAC/DBSCAN segmentation, point-to-plane ICP registration, and constrained trajectory optimization with CasADi/IPOPT, with checks of kinematics, joint limits, and path constraints in MuJoCo.

Future Research

I would build on these results by studying how evolving pose and contact estimates should change manipulation actions, when failure analysis identifies a need for additional demonstrations, and how learned grasp proposals can be combined with constrained feedback execution. Controlled changes in sensing, contact conditions, and execution timing would separate errors in perception, action selection, and control. Repeatable comparisons with unchanged baselines would test whether each intervention improves task completion while preserving the applicable safety constraints.

Selected References

[1] C. Enwerem et al. *Variational Neural Belief Parameterizations for Robust Dexterous Grasping under Multimodal Uncertainty*. IROS 2026. arXiv:2604.25897.

[2] C. Enwerem et al. *FIRMGrasp: A Friction-Informed Risk Margin for Robust Grasp Synthesis*. Preprint, 2026. arXiv:2607.25049.

[3] C. Enwerem et al. *Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality*. Preprint, 2026. arXiv:2608.00600.

[4] C. Enwerem et al. *Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds*. Preprint, 2026. arXiv:2609.01453, project page.

[5] C. Enwerem et al. *EquiDexFlow: Contact-Grounded SE(3)-Equivariant Dexterous Grasp Generative Flows*. Preprint, 2026. arXiv:2606.12728.

[6] C. Enwerem, S. Kemal, J. S. Baras, and C. Belta. *Belief-Informed Hybrid Control with Almost-Sure Target-Set Convergence*. Preprint, 2026. arXiv:2610.07674.

[7] C. Enwerem, A. G. Puranic, J. S. Baras, and C. Belta. *Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression*. CDC, 2025. doi:10.1109/CDC57313.2025.11312575; arXiv:2506.06954.